HEIGHTS ON PROJECTIVE SPACE

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To apply the descent theorem to $E(K)$, one needs a height on $E(K)$ with the Northcott property. Silverman builds it from a height on projective space [Sil09, §VIII.5]: a point of $\mathbb{P}^N(\overline{\mathbb{Q}})$ is large when its coordinates are large at the archimedean places or have large denominators at the non-archimedean ones. This worksheet works out the first properties of this height. Problem 1.5, the heart of the worksheet, proves Kronecker's theorem: the algebraic numbers of height 1 are 0 and the roots of unity. Problem 2.1 proves the inequalities for products and sums, and the behavior of the height under the Segre and d-uple embeddings. Problem 3.2 computes the height through the ideal generated by the coordinates, and shows on $\mathbb{Q}(\sqrt{-5})$ why the recipe of coprime integer coordinates does not extend to every number field. Each problem is preceded by the definitions and results it uses.

Standing notation.

- An algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$, and its subring $\overline{\mathbb{Z}}$ of algebraic integers.
- A number field K, its ring of integers $\mathcal{O}_K$, its set of places $\mathcal{P}_K$, and the subsets $\mathcal{P}_K^\infty$ and $\mathcal{P}_K^0$ of archimedean and non-archimedean places.
- For $v \in \mathcal{P}_K$, the completion K_v, the absolute value $|\ |_v$ on K_v extending the usual absolute value on $\mathbb{Q}_p$ or $\mathbb{R}$, and the local degree $n_v \coloneqq [K_v : \mathbb{Q}_v]$, where $\mathbb{Q}_v$ is the completion of $\mathbb{Q}$ at the restriction of v.
- For a point $P = [x_0 : \cdots : x_N]$ of $\mathbb{P}^N(K)$ and $v \in \mathcal{P}_K$, the number $|P|_v \coloneqq \max_i |x_i|_v$. It depends on the choice of coordinates.

How to use this. Allow about 25 minutes for Problem 1.5, 15 for Problem 2.1 and 20 for Problem 3.2. Problem 1.5 is the one not to skip. The `Magma` blocks run in seconds in the free online calculator. Everything else is by hand.

1. Kronecker's theorem

This section is [Sil09, Exercise 8.6], with the proof of [HS00, Corollary B.2.3.1]. The definitions and the first properties of the height are [Sil09, §VIII.5]. The last two items compare Kronecker's theorem with the enumeration of polynomials that proves it in small degree, and with algebraic numbers of small height.

Proposition 1.1 (Extension and product formulas [Sil09, (VIII.5.2), (VIII.5.3)])**.** *(1) Let L be a finite extension of K, and let $v \in \mathcal{P}_K$. Then $\sum_{w|v} n_w = [L : K]\, n_v$, the sum over the places $w \in \mathcal{P}_L$ that restrict to v.*
(2) Let $x \in K^\times$. Then $\prod_{v \in \mathcal{P}_K} |x|_v^{n_v} = 1$.

Definition 1.2 (Heights on projective space [Sil09, §§VIII.5, VIII.6])**.** The height relative to K of a point $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(K)$ is

$$H_K(P) \coloneqq \prod_{v \in \mathcal{P}_K} |P|_v^{n_v}.$$

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Let $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$, and let K be a number field with $P \in \mathbb{P}^N(K)$. The height of P is $H(P) := H_K(P)^{1/[K:\mathbb{Q}]}$, and its logarithmic height is $h(P) := \log H(P)$. For $x \in \overline{\mathbb{Q}}$, let $H(x) := H([x:1])$ and $h(x) := \log H(x)$.

Proposition 1.3 (Properties of the height [Sil09, Proposition VIII.5.4]). *Let $P \in \mathbb{P}^N(K)$.*

(1) The height $H_K(P)$ does not depend on the choice of coordinates of P.
(2) The height satisfies $H_K(P) \geqslant 1$.
(3) Let L be a finite extension of K. Then $H_L(P) = H_K(P)^{[L:K]}$.

In particular, $H(P)$ does not depend on the choice of K, and $H(P) \geqslant 1$.

Theorem 1.4 (Northcott's theorem [Sil09, Theorem VIII.5.11]). *Let C and d be real numbers. Then the set of points $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$ with $H(P) \leqslant C$ and $[\mathbb{Q}(P) : \mathbb{Q}] \leqslant d$ is finite, where $\mathbb{Q}(P)$ is the field generated by the ratios x_i/x_j with $x_j \neq 0$ of the coordinates of P. In particular, the set $\{P \in \mathbb{P}^N(K) : H_K(P) \leqslant C\}$ is finite.*

Problem 1.5 (Kronecker's theorem).
This problem proves Kronecker's theorem: a nonzero algebraic number $x \in \overline{\mathbb{Q}}^\times$ has $H(x) = 1$ if and only if x is a root of unity. By (2), it is equivalent to Kronecker's original form: a nonzero algebraic integer $x \in \overline{\mathbb{Z}}$ with $|\tau(x)| \leqslant 1$ for every embedding $\tau\colon \mathbb{Q}(x) \to \mathbb{C}$ is a root of unity.
(1) Show that every root of unity has height 1. Show that $H(0) = 1$, so the hypothesis $x \neq 0$ cannot be dropped.
(2) Let $x \in \overline{\mathbb{Q}}$ and $K := \mathbb{Q}(x)$. Show that $H(x) = 1$ if and only if $|x|_v \leqslant 1$ for every $v \in \mathcal{P}_K$, and that this holds if and only if $x \in \overline{\mathbb{Z}}$ and $|\tau(x)| \leqslant 1$ for every embedding $\tau\colon K \to \mathbb{C}$. Show that $x = (3+4i)/5$ satisfies $|\tau(x)| = 1$ for both embeddings of $\mathbb{Q}(i)$, and compute $H(x)$.
(3) Let $x \in \overline{\mathbb{Q}}$. Show that $H(x^n) = H(x)^n$ for every integer $n \geqslant 1$, and that $H(1/x) = H(x)$ if $x \neq 0$. Prove Kronecker's theorem.[a]
(4) Let $f = t^d + a_1 t^{d-1} + \cdots + a_d \in \mathbb{Z}[t]$ be monic of degree $d \geqslant 1$, with all complex roots in the closed unit disc. Show that $|a_i| \leqslant \binom{d}{i}$ for every i.[b] So there are finitely many such f of each degree. `Magma` lists those of degree 4 with $a_4 \neq 0$, and checks that each is a product of cyclotomic polynomials:

```
R<t> := PolynomialRing(Integers());
d := 4;
B := [[-Binomial(d, i)..Binomial(d, i)] : i in [1..d]];
S := [];
for c in CartesianProduct(B) do
  f := t^d + &+[c[i]*t^(d - i) : i in [1..d]];
  if c[d] ne 0 and forall{r : r in Roots(f, ComplexField(30)) |
                          Abs(r[1]) le 1 + 10^-10} then
    Append(~S, f);
  end if;
end for;
#CartesianProduct(B), #S;                          // 3159 24
forall{f : f in S | forall{g : g in Factorization(f) |
                           IsCyclotomicPolynomial(g[1])}};        // true
```

Use this output to prove Kronecker's theorem for every $x \in \overline{\mathbb{Q}}^\times$ with $[\mathbb{Q}(x) : \mathbb{Q}] \leqslant 4$, without Theorem 1.4.

(5) Let $d \geqslant 1$, and let $2^{1/d}$ be the real root of $t^d - 2$. Show that $h(2^{1/d}) = (\log 2)/d$. So h takes arbitrarily small positive values on $\overline{\mathbb{Q}}^\times$. `Magma` checks this, and computes the logarithmic height of a root b of Lehmer's polynomial:

```
R<t> := PolynomialRing(Integers());
[RealField(10) | d*AbsoluteLogarithmicHeight(NumberField(t^d - 2).1)
                 : d in [2..5]];       // [ 0.6931471805, 0.6931471805, ... ]
RealField(10)!Log(2);                  // 0.6931471805
f := t^10 + t^9 - t^7 - t^6 - t^5 - t^4 - t^3 + t + 1;
IsIrreducible(f), IsCyclotomicPolynomial(f);          // true false
K<b> := NumberField(f);
RealField(10)!AbsoluteLogarithmicHeight(b);           // 0.01623576120
```

Explain why $h(b) > 0$, and compute $[\mathbb{Q}(b) : \mathbb{Q}]\, h(b)$.

[a] *Hint:* if $H(x) = 1$, consider the set $\{x, x^2, x^3, \dots\}$.
[b] *Hint:* the coefficient a_i is, up to sign, the i-th elementary symmetric function of the roots.

2. Basic properties of the height

This section is [Sil09, Exercise 8.8]. The last two items are [HS00, Proposition B.2.4(b), (c)], for the Segre embedding [Har77, Exercise I.2.14] and the d-uple embedding [Har77, Exercise I.2.12]. The heights are those of Definition 1.2.

Problem 2.1 (Basic properties of the height)**.**
In (1) and (2), let $N \geqslant 1$ and $x_1, \dots, x_N \in \overline{\mathbb{Q}}$.
(1) Show that $H(x_1 x_2 \cdots x_N) \leqslant H(x_1)H(x_2)\cdots H(x_N)$. Show that the inequality can be strict.
(2) Show that $H(x_1 + x_2 + \cdots + x_N) \leqslant N\, H(x_1)H(x_2)\cdots H(x_N)$, and that the factor N cannot be replaced by a smaller one.[a]
(3) Let $N, M \geqslant 0$, and let $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(\overline{\mathbb{Q}})$ and $Q = [y_0 : \cdots : y_M] \in \mathbb{P}^M(\overline{\mathbb{Q}})$. The Segre embedding sends (P, Q) to the point

$$S(P, Q) := [x_0 y_0 : x_0 y_1 : \cdots : x_i y_j : \cdots : x_N y_M] \in \mathbb{P}^{MN+M+N}(\overline{\mathbb{Q}}),$$

with the products $x_i y_j$ in lexicographic order of (i, j). Show that $S(P, Q)$ does not depend on the choice of coordinates, and that $H(S(P, Q)) = H(P)H(Q)$.
(4) Let $N \geqslant 0$ and $d \geqslant 1$, let $M := \binom{N+d}{N} - 1$, and let $f_0, \dots, f_M$ be the distinct monomials of degree d in $X_0, \dots, X_N$. The d-uple embedding sends $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(\overline{\mathbb{Q}})$ to $P^{(d)} := [f_0(P) : \cdots : f_M(P)] \in \mathbb{P}^M(\overline{\mathbb{Q}})$. Show that

$$H(P^{(d)}) = H(P)^d = H([x_0^d : \cdots : x_N^d]).$$

[a] *Hint:* the triangle inequality reads $|t_1 + \cdots + t_N|_v \leqslant N^{\epsilon(v)} \max_i |t_i|_v$, with $\epsilon(v) = 1$ if v is archimedean and $\epsilon(v) = 0$ otherwise.

3. Heights and ideals

This section is [Sil09, Exercise 8.9], which extends [Sil09, Remark VIII.5.5] from $\mathbb{Q}$ to a number field. The description of the non-archimedean places is [HS00, Proposition B.1.3] and the discussion before it. For a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$, let $\mathrm{ord}_\mathfrak{p} \colon K^\times \to \mathbb{Z}$ be the exponent of $\mathfrak{p}$ in the

factorization of the fractional ideal $x\mathcal{O}_K$, and $\operatorname{Nm}\mathfrak{p} \coloneqq \#\mathcal{O}_K/\mathfrak{p}$. The norm extends multiplicatively to fractional ideals: $\operatorname{Nm}_{K/\mathbb{Q}}\mathfrak{a} \coloneqq \prod_{\mathfrak{p}}(\operatorname{Nm}\mathfrak{p})^{\operatorname{ord}_{\mathfrak{p}}\mathfrak{a}}$.

Proposition 3.1 (Non-archimedean places [HS00, Proposition B.1.3])**.** *The non-archimedean places of K correspond to the nonzero prime ideals of $\mathcal{O}_K$. If $v \in \mathcal{P}_K^0$ corresponds to $\mathfrak{p}$, then $|x|_v^{n_v} = (\operatorname{Nm}\mathfrak{p})^{-\operatorname{ord}_{\mathfrak{p}}(x)}$ for every $x \in K^\times$.*

Problem 3.2 (Heights and ideals)**.**

(1) Let $x_0, \dots, x_N \in K$, not all zero, let $P \coloneqq [x_0 : \cdots : x_N] \in \mathbb{P}^N(K)$, and let $\mathfrak{b}$ be the fractional ideal of K generated by $x_0, \dots, x_N$. Show that

$$H_K(P) = \left(\operatorname{Nm}_{K/\mathbb{Q}}\mathfrak{b}\right)^{-1} \prod_{v \in \mathcal{P}_K^\infty} |P|_v^{n_v}.$$

(2) Let $K = \mathbb{Q}$. Deduce that if $x_0, \dots, x_N$ are coprime integers, then $H_{\mathbb{Q}}(P) = \max_i |x_i|$. More generally, show that if $\mathfrak{b}$ is principal, then P has coordinates in $\mathcal{O}_K$ that generate the unit ideal, and for such coordinates $H_K(P) = \prod_{v \in \mathcal{P}_K^\infty} |P|_v^{n_v}$.

(3) Let $K \coloneqq \mathbb{Q}(\sqrt{-5})$, with $a \coloneqq \sqrt{-5}$, and let $P \coloneqq [2 : 1 + a] \in \mathbb{P}^1(K)$. `Magma` computes both sides of (1), and compares with its own logarithmic height of $(1 + a)/2$:

```
K<a> := QuadraticField(-5);
O := Integers(K);
xs := [K | 2, 1 + a];
b := ideal<O | xs>;
IsPrincipal(b), Norm(b);                                    // false 2
arch := &*[Max([Abs(Evaluate(x, v)) : x in xs])^LocalDegree(v)
          : v in InfinitePlaces(K)];
primes := &join[Support(x*O) : x in xs | x ne 0];
fin := &*[Norm(p)^(-Min([Valuation(x, p) : x in xs])) : p in primes];
RealField(10)!arch, fin;                                    // 6.000000000 1/2
[RealField(10) | AbsoluteLogarithmicHeight((1 + a)/2), Log(3)/2];
                                          // [ 0.5493061444, 0.5493061444 ]
```

Check the output by hand: show that $\mathfrak{b}$ has norm 2 and is not principal, and compute $H_K(P)$ and $H(P)$. Show that P has no coordinates in $\mathcal{O}_K$ that generate the unit ideal. Explain the last line of the output.

References

[Har77] Robin Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, vol. No. 52, Springer-Verlag, New York-Heidelberg, 1977. MR463157 ↑

[HS00] Marc Hindry and Joseph H. Silverman, *Diophantine geometry*, Graduate Texts in Mathematics, vol. 201, Springer-Verlag, New York, 2000. An introduction. MR1745599 ↑

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑